Problem 10.1

A deposit of $500 is invested at time 5 years. The constant force of interest is 6% per year. Determine the accumulated value of the investment at the end of 10 years.

Solution

The deposit is made at time 5 and is accumulated to time 10, so it earns interest for

$$10 - 5 = 5$$

years.

With a constant force of interest $\delta = 0.06$, the equivalent annual effective interest rate i satisfies

$$\delta = \ln(1+i).$$

Thus,

$$0.06 = \ln(1+i),$$

so

$$1 + i = e^{0.06} \qquad \Longrightarrow \qquad i = e^{0.06} - 1.$$

The accumulated value after 5 years is

$$500(1+i)^5.$$

Since $1 + i = e^{0.06}$,

$$500(1+i)^5 = 500(e^{0.06})^5 = 500e^{0.30}.$$

Therefore,

$$500e^{0.30} = 674.9294038.$$

Thus, the accumulated value is

$$\boxed{674.93}.$$